Yuhao Bao

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RESEARCH INTERESTS

Human-centered AI reliability and evaluation: how vision-language and large language models interpret crisis-related social media, how their outputs compare with human annotations, and when their judgments should not be trusted in high-stakes settings.

EDUCATION

Brigham Young University, Provo, UT — Expected April 2027
B.S. in Computer Science; Minors in Mathematics and Physics
GPA: 3.85/4.00 (Computer Science: 3.90/4.00)
Relevant coursework: Advanced Deep Learning; Multi-Agent Systems; Database Modeling

PUBLICATIONS AND MANUSCRIPTS

Bao, Y., & Hughes, A. (2026). When Social Media Images Need Words: Measuring Context Gap and Fusion Tax in Crisis Image Captioning. *Proceedings of the International ISCRAM Conference, 23*. Work-in-Progress paper. doi:10.59297/4bar0c32

Bao, Y. What Agreement Hides: LLM Labeling of Hurricane Dorian Tweets (working title). In preparation for the International ISCRAM Conference 2027.

RESEARCH EXPERIENCE

Research Assistant, HCI Research Group, Brigham Young University — Nov 2023 – Present
Advisor: Prof. Amanda Hughes

LLM labeling of Hurricane Dorian tweets (first-author manuscript, in preparation)

- Compared five model versions from three families (Qwen3.8-27B, Gemini 3.7 Flash, Gemini 3.8 Flash, GPT-6 Sol, GPT-6.1 Sol) on 355 Hurricane Dorian tweets with 433 photos against historical TREC-IS Information Type and Priority labels, using identical inputs and a fixed prompt.
- Found that headline agreement hides systematic differences: Information Type partial overlap was 79.6%–85.8% while exact set agreement was 1.1%–4.2%; Priority agreement ranged from 39.2% to 50.1%, and GPT-6 Sol assigned Medium to 77.2% of tweets; all five rated the three Critical-reference tweets as Low or Medium (agreement with historical labels, not correctness).

Crisis image captioning with vision-language models (ISCRAM 2026)

- Compared three vision-language models (Gemini 2.0 Flash, Qwen2.5-VL, BLIP) on 204 CrisisFACTS image-post pairs under image-only and image-plus-text conditions.
- Designed the scoring rubric and error taxonomy and carried out the human coding (Krippendorff's $\alpha = 0.69$).
- Built Python batch evaluation pipelines with JSON parsing and rubric-based scoring; adding post text raised Gemini captioning accuracy from 75.5% to 91.7% and reduced object misidentification from 15.7% to 2.9%.

Research Project: Text-to-SQL with Retrieval and LoRA Fall 2025 – Winter 2026

Brigham Young University. Advisor: Prof. Porter Jenkins

- Built a Text-to-SQL RAG pipeline on Qwen3.5-9B with BM25 top-5 retrieval, fine-tuned on 6,169 examples and validated on 1,089.
- Implemented Shadow-LoRA (Base-to-Instruct adapter transfer) against a matched Direct-LoRA baseline; exact match rose from 19.47% to 30.49% and LLM-judge pass rate from 56.03% to 58.53%. Analyzed a reproducibility gap against the historical best (37.28% exact match).

OPEN-SOURCE PROJECTS

modelagree (GitHub): open-source tool for measuring agreement between LLM labels, with 107 automated tests; under development.

crisis-tweet-classifier (GitHub): compared TF-IDF with logistic regression, fine-tuned DistilBERT, and a zero-shot local LLM (qwen2.5:7b) on the HumAID crisis-tweet dataset with an event-wise split, reporting macro-F1, calibration (ECE), and inference cost; on the same 300 test tweets, DistilBERT reached macro-F1 0.78 versus 0.61 zero-shot.

INDUSTRY EXPERIENCE

Software Engineering Intern, Jingus (J.ing), Guangzhou, China May 2023 – Aug 2023

- Developed Shopify Liquid theme components for an apparel storefront and integrated the PayPal payment gateway via REST API and webhooks to automate order status sync.

SKILLS

Research methods: rubric and error-taxonomy design, qualitative coding, inter-rater reliability, LLM and VLM evaluation
Programming: Python, SQL, Java, C++, JavaScript/TypeScript
ML and data: PyTorch, Hugging Face Transformers, LoRA/PEFT, BM25 retrieval, pandas, NumPy, PostgreSQL, Git, Linux

HONORS AND AFFILIATIONS

BYU travel funding to attend the 2024 SHPE National Convention (selected, limited awards)
Member: SHPE, ACM, BYU AI Club